

AVALON, Australia

WMO: 948540

Lat: 38.0286S

Lon: 144.4783E

Elev: 11

StdP: 101.19

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.4	2.9	-0.1	3.8	8.2	1.1	4.1	8.2	14.2	11.9	13.1	12.3	2.0	300	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.8	35.5	19.5	32.2	18.8	29.1	18.2	21.6	28.7	20.6	27.7	19.7	26.4	7.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.5	23.0	18.5	13.4	22.3	17.7	12.7	21.7	62.9	28.7	59.1	27.7	56.2	26.5	33.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.4	10.3	DB	-1.3	41.6	1.0	2.5	-2.0	43.4	-2.6	44.9	-3.2	46.3	-3.9	48.1	(4)
(5)			WB	-1.5	25.2	1.0	3.4	-2.2	27.7	-2.7	29.7	-3.3	31.6	-4.0	34.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	19.9	20.0	18.3	15.1	12.4	10.1	9.7	10.2	11.9	13.9	16.1	18.0	(6)	
	DBStd	4.72	3.89	3.45	3.37	2.82	2.45	2.14	2.01	2.19	2.63	3.20	3.53	3.72	(7)	
	HDD10.0	91	0	0	0	1	5	24	30	23	7	2	0	0	(8)	
	HDD18.3	1619	23	18	43	106	184	246	267	251	194	146	88	53	(9)	
	CDD10.0	1767	307	279	256	152	80	28	21	30	64	121	182	247	(10)	
	CDD18.3	253	71	64	41	8	1	0	0	0	1	7	21	41	(11)	
	CDH23.3	2708	704	549	395	99	3	0	0	1	20	163	276	498	(12)	
(13)	CDH26.7	1264	373	252	171	27	0	0	0	0	3	58	117	262	(13)	
(14) Wind	WSAvg	5.1	5.6	5.4	5.1	4.5	4.7	4.4	5.0	5.3	5.3	5.3	5.3	5.5	(14)	
(15) Precipitation	PrecAvg	529	39	39	31	43	37	50	47	49	52	50	54	38	(15)	
	PrecMax	719	116	129	85	163	77	107	89	78	98	118	111	113	(16)	
	PrecMin	312	2	2	10	9	11	22	18	22	22	8	15	2	(17)	
	PrecStd	102	27	29	17	30	20	21	20	16	21	30	29	25	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	36.9	36.1	30.0	23.2	18.2	18.0	21.1	25.9	32.2	33.9	38.3	(19)	
		MCWB	20.6	19.7	19.8	16.9	14.4	12.8	11.8	13.6	15.0	16.5	18.4	19.3	(20)	
	2%	DB	35.4	33.2	31.1	26.1	20.8	16.2	16.1	18.1	22.2	27.7	30.2	33.8	(21)	
		MCWB	19.9	19.4	18.7	15.6	14.0	12.2	11.4	12.0	13.9	15.4	18.3	19.0	(22)	
	5%	DB	30.3	29.8	27.6	23.0	18.8	15.1	14.8	16.2	19.8	24.1	26.2	28.9	(23)	
		MCWB	19.4	19.1	17.9	15.5	13.4	11.7	10.7	11.4	13.4	15.0	17.1	18.3	(24)	
	10%	DB	26.2	26.2	24.2	20.8	17.1	14.1	13.8	14.9	17.6	20.9	22.9	24.8	(25)	
		MCWB	19.0	18.9	17.5	15.0	13.0	11.2	10.3	10.7	12.4	14.1	16.4	17.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.8	22.2	21.3	18.5	16.2	14.1	13.2	14.3	16.2	18.6	21.1	22.7	(27)	
		MCDB	30.3	28.6	28.9	25.7	19.8	16.4	16.0	19.1	22.5	25.7	28.8	30.4	(28)	
	2%	WB	21.7	21.1	19.9	17.2	15.0	13.1	12.1	12.8	14.9	16.8	19.3	20.3	(29)	
		MCDB	28.9	27.8	26.7	22.0	19.1	15.2	14.8	16.8	20.2	24.0	26.3	28.5	(30)	
	5%	WB	20.5	20.3	19.0	16.3	14.1	12.3	11.2	11.8	13.7	15.6	18.0	19.1	(31)	
		MCDB	28.0	26.5	25.0	21.0	17.6	14.6	13.9	15.4	18.6	22.4	24.3	26.3	(32)	
	10%	WB	19.3	19.5	18.1	15.5	13.3	11.5	10.6	11.0	12.7	14.5	16.8	18.0	(33)	
		MCDB	26.2	25.3	23.2	19.8	16.4	13.8	13.2	14.2	16.9	20.1	21.8	24.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	10.8	10.7	10.2	8.8	8.0	8.1	9.0	10.3	11.7	11.2	11.7	(35)	
		MCDBR	20.0	18.0	17.3	15.3	11.5	8.9	9.3	11.4	14.3	18.8	18.2	20.3	(36)	
	5% WB	MCWBR	7.0	6.4	6.9	7.0	6.4	5.8	5.7	6.7	7.5	8.6	7.8	7.6	(37)	
		MCWBR	15.6	13.3	13.6	12.1	9.8	8.4	8.4	10.2	12.5	15.5	14.9	15.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.353	0.356	0.334	0.328	0.315	0.302	0.303	0.313	0.334	0.340	0.349	0.345	(40)	
		taud	2.497	2.498	2.559	2.549	2.566	2.591	2.580	2.543	2.475	2.474	2.478	2.510	(41)	
	Ebn at Noon		979	952	934	876	828	812	833	880	918	957	977	992	(42)	
		Edh at Noon	113	108	94	83	71	64	68	82	100	110	115	113	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.12	6.22	4.92	3.42	2.29	1.79	2.01	2.83	4.05	5.38	6.27	7.06	(44)	
	RadStd		0.44	0.29	0.28	0.19	0.12	0.13	0.11	0.17	0.26	0.40	0.37	0.40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		+0.50	N/A	N/A	+0.88	N/A	N/A	-17	-129	+175	+59

Nomenclature: See separate page